1. For the equation of the Cobb-Douglas production function:

$$Q = \lambda K^{\beta_1} L^{\beta_2} M^{\beta_3} e^u, \tag{1}$$

we can take the log on both sides of the equation:

$$\begin{aligned} log(Q) &= log(\lambda K^{\beta_1} L^{\beta_2} M^{\beta_3} e^u) \\ &= log(\lambda) + log(K^{\beta_1}) + log(L^{\beta_2}) + log(M^{\beta_3}) + log(e^u) \\ &= log(\lambda) + \beta_1 log(K) + \beta_2 log(L) + \beta_3 log(M) + u \\ &= \beta_0 + \beta_1 log(K) + \beta_2 log(L) + \beta_3 log(M) + u, \end{aligned} \tag{2}$$

where we define $\beta_0 = log(\lambda)$. Therefore, by taking the log on both sides, we obtain a log-log model, and the parameters can be estimated by OLS. Given that it is a log-log model, the interpretation of β_1 is that (keeping other explanatory variables fixed/unchanged)

$$\beta_1 \approx \frac{\Delta Q/Q}{\Delta K/K} = \frac{\text{percentage change in } Q}{\text{percentage change in } K}, \tag{3}$$

so that if we increase K (capital) by 1%, the corresponding change in Q (production) is approximately $\beta_1 \times 1\% = \beta_1\%$.
Similarly, the interpretation of β_2 is that (keeping other explanatory variables fixed/unchanged)

$$\beta_2 \approx \frac{\Delta Q/Q}{\Delta L/L} = \frac{\text{percentage change in } Q}{\text{percentage change in } L}, \tag{4}$$

so that if we increase L (labour) by 1%, the corresponding change in Q (production) is approximately $\beta_2 \times 1\% = \beta_2\%$.

2. (a) Notice that the model can be written as:

$$log(Earnings) = \beta_0 + \beta_1 Female + \beta_2 log(MarketValue) + \beta_3 Return + u, \tag{5}$$

and the OLS gives:

$$log(\widehat{Earnings}) = \hat\beta_0 + \hat\beta_1 Female + \hat\beta_2 log(MarketValue) + \hat\beta_3 Return. \tag{6}$$

Therefore, according to the question, $\hat\beta_1 = -0.28$ and $\hat\beta_2 = 0.37$.
For β_1, notice that it corresponds to a log-linear model, so that (keeping other explanatory variables unchanged)

$$\begin{aligned} \beta_1 &\approx \frac{\Delta Earnings/Earnings}{\Delta Female}, \\ \Delta Earnings/Earnings &\approx \beta_1 \cdot \Delta Female, \end{aligned} \tag{7}$$

Similarly, for the OLS estimator $\hat{\beta}_1$, we have (keeping other explanatory variables unchanged)

$$\begin{aligned} \hat{\beta}_1 &\approx \frac{\Delta \widehat{Earnings/Earnings}}{\Delta Female}, \\ \Delta \widehat{Earnings/Earnings} &\approx \hat{\beta}_1 \cdot \Delta Female. \end{aligned} \tag{8}$$

Therefore, when $\Delta Female = 1$ (changing from $Female = 0$ to $Female = 1$), the corresponding percentage change in earnings predicted by the OLS is -28%, which means that female earning is lower than male earning by 28%:

$$\Delta \widehat{Earnings/Earnings} \approx \hat{\beta}_1 \cdot \Delta Female = -0.28 \cdot 1 = -0.28 = -28\%. \tag{9}$$

(b) For β_2, notice that it corresponds to a log-log model, so that (keeping other explanatory variables unchanged)

$$\begin{aligned} \beta_2 &\approx \frac{\Delta Earnings/Earnings}{\Delta MarketValue/MarketValue}, \\ \Delta Earnings/Earnings &\approx \beta_2 \cdot \Delta MarketValue/MarketValue, \end{aligned} \tag{10}$$

Similarly, for the OLS estimator $\hat{\beta}_2$, we have (keeping other explanatory variables unchanged)

$$\begin{aligned} \hat{\beta}_2 &\approx \frac{\Delta \widehat{Earnings/Earnings}}{\Delta MarketValue/MarketValue}, \\ \Delta \widehat{Earnings/Earnings} &\approx \hat{\beta}_2 \cdot \Delta MarketValue/MarketValue. \end{aligned} \tag{11}$$

Therefore, when $\Delta MarketValue/MarketValue = 1\%$ (the percentage change of the market value is 1%), the corresponding predicted percentage change in earnings is

$$\Delta \widehat{Earnings/Earnings} \approx \hat{\beta}_2 \cdot 1\% = 0.37 \cdot 1\% = 0.37\%. \tag{12}$$

3. (a) We can define 3 categories totally: $\{STR < 20\}$, $\{20 \leq STR < 25\}$, and $\{25 \leq STR\}$. Therefore, we should use 2 dummy variables to avoid multicollinearity.

(b) For example, let us define 2 dummy variables STR_{small} and STR_{medium} as follows:
$STR_{small,i}$ equals 1 if $STR_i < 20$ and equals 0 otherwise;
$STR_{medium,i}$ equals 1 if $20 \leq STR_i < 25$ and equals 0 otherwise.
Then, the linear regression model can be defined as follows:

$$TestScore_i = \beta_0 + \beta_1 STR_{small,i} + \beta_2 STR_{medium,i} + u_i, \quad i = 1, ..., n. \tag{13}$$

The interpretation of the model is as follows (given that the OLS Assumption 1 holds):

$$\begin{aligned} E[TestScore_i | STR_{small,i} = 0,\ STR_{medium,i} = 0] &= E[TestScore_i | STR_i \geq 25] = \beta_0, \\ E[TestScore_i | STR_{small,i} = 1,\ STR_{medium,i} = 0] &= E[TestScore_i | STR_i < 20] = \beta_0 + \beta_1, \\ E[TestScore_i | STR_{small,i} = 0,\ STR_{medium,i} = 1] &= E[TestScore_i | 20 \leq STR_i < 25] = \beta_0 + \beta_2, \end{aligned}$$

which also shows the interpretation of β_1 and β_2:

$$\begin{aligned}\beta_1 &= E[TestScore_i|STR_i < 20] - E[TestScore_i|STR_i \geq 25] \\ \beta_2 &= E[TestScore_i|20 \leq STR_i < 25] - E[TestScore_i|STR_i \geq 25].\end{aligned}$$

(c) The relation between STR and $TestScore$ will be three steps: one flat step for $\{STR < 20\}$ (this one is the highest), one flat step for $\{20 \leq STR < 25\}$ (this one is in the middle), and one flat step for $\{25 \leq STR\}$ (this one is the lowest).